

Problem

Let n and k be positive integers with $n > k$. Prove that

$$\frac{1}{n+1} \cdot \frac{n^n}{k^k(n-k)^{n-k}} < \frac{n!}{k!(n-k)!} < \frac{n^n}{k^k(n-k)^{n-k}}.$$

Source: APMO 1998

§1 Extracting the Hidden Binomial Structure

Multiplying the desired inequality by $k^k(n-k)^{n-k}$, we see that it is equivalent to

$$\frac{n^n}{n+1} < \binom{n}{k} k^k (n-k)^{n-k} < n^n.$$

The appearance of the quantity

$$\binom{n}{k} k^k (n-k)^{n-k}$$

suggests considering the binomial expansion

$$n^n = ((n-k) + k)^n.$$

Expanding, we obtain

$$n^n = \sum_{i=0}^n \binom{n}{i} (n-k)^{n-i} k^i.$$

Let

$$T_{i+1} = \binom{n}{i} (n-k)^{n-i} k^i, \quad 0 \leq i \leq n.$$

Then

$$n^n = T_1 + T_2 + \cdots + T_{n+1}.$$

Observe that

$$T_{k+1} = \binom{n}{k} k^k (n-k)^{n-k}.$$

Thus the quantity appearing in the statement is precisely the $(k+1)$ st term of the expansion.

§2 Locating the Largest Term

To understand the size of T_{k+1} , consider the ratio of consecutive terms:

$$\frac{T_{i+1}}{T_i} = \frac{\binom{n}{i}(n-k)^{n-i}k^i}{\binom{n}{i-1}(n-k)^{n-i+1}k^{i-1}} = \frac{(n-i+1)k}{i(n-k)}.$$

Hence

$$\begin{aligned} \frac{T_{i+1}}{T_i} > 1 &\iff (n-i+1)k > i(n-k) \\ &\iff nk + k > in \\ &\iff i < k + \frac{k}{n}. \end{aligned}$$

Since $0 < k < n$, we have

$$0 < \frac{k}{n} < 1,$$

and therefore

$$\frac{T_{i+1}}{T_i} > 1 \iff i \leq k.$$

Consequently,

$$T_1 < T_2 < \cdots < T_{k+1} > T_{k+2} > \cdots > T_{n+1}.$$

Thus T_{k+1} is the largest term in the entire expansion.

§3 Bounding the Central Term

Since all terms of the expansion are positive,

$$T_{k+1} < T_1 + \cdots + T_{n+1} = n^n.$$

Therefore

$$\binom{n}{k} k^k (n-k)^{n-k} < n^n,$$

which yields

$$\frac{n!}{k!(n-k)!} < \frac{n^n}{k^k(n-k)^{n-k}}.$$

For the lower bound, note that the largest term of a collection of $n+1$ positive numbers is at least their average. Hence

$$T_{k+1} > \frac{T_1 + \cdots + T_{n+1}}{n+1} = \frac{n^n}{n+1}.$$

Therefore

$$\binom{n}{k} k^k (n-k)^{n-k} > \frac{n^n}{n+1},$$

which implies

$$\frac{n!}{k!(n-k)!} > \frac{1}{n+1} \cdot \frac{n^n}{k^k(n-k)^{n-k}}.$$

Combining the two inequalities gives

$$\boxed{\frac{1}{n+1} \cdot \frac{n^n}{k^k(n-k)^{n-k}} < \frac{n!}{k!(n-k)!} < \frac{n^n}{k^k(n-k)^{n-k}}}.$$

§4 A Probabilistic Perspective

Dividing the inequality by n^n gives

$$\frac{1}{n+1} < \binom{n}{k} \left(\frac{k}{n}\right)^k \left(1 - \frac{k}{n}\right)^{n-k} < 1.$$

The middle expression is exactly the probability that a binomial random variable with parameters

$$n \quad \text{and} \quad p = \frac{k}{n}$$

takes the value k . The proof above shows that this probability is the maximal term in the corresponding binomial distribution, thereby recovering the same inequalities from a probabilistic viewpoint.